[LUP-2] Gold volume-surge deep dive: lookback, ratio, velocity, time of day, and GC vs MGC lead/lag Created: 20/Aug/26 Updated: 21/Aug/26

Status:	To Do
Project:	Lupine Research
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Task	**Priority:**	Medium
Reporter:	Adam Balter	**Assignee:**	Unassigned
Resolution:	Unresolved	**Votes:**	0
Labels:	Strat4aGold, microstructure, volume-surge		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Rank:	0|i001pz:

Description

Objective

Determine whether short bursts of trading volume in gold futures predict short-horizon moves in the gold ETF **GLD**, and if so, specify the best measurement of that effect.

Earlier research found that a simple volume-surge construction performed well near the US cash close (approximately 15:00–16:00 Eastern). This ticket is a structured re-examination of that hypothesis: test alternative definitions, quantify robustness, and recommend whether to retain, revise, or replace the current specification.

Internal strategy labels from prior packs are optional context only. The work below is defined entirely in market and measurement terms.

Background

Instruments

Name	Description	Role
GLD	Gold ETF listed on US equities	Primary prediction / trading target
GC	Full-size COMEX gold futures	Primary futures volume source; often leads price discovery
MGC	Micro gold futures	Related contract with different liquidity and timing
Related gold ETFs (GLDM, IAU, IAUM)	Additional gold ETF complex	Optional activity aggregation with GLD

Economic rationale: Futures markets can exhibit volume and aggression spikes before that pressure is fully reflected in the ETF. The task is to measure those spikes in a way that is predictive, stable, and tradeable.

Baseline volume-surge definition

On **1-minute bars**, the specification used previously was:

1. Current-minute traded volume
2. Divided by average volume over the prior **30 minutes** (30 bars)
3. Divided again by an **expanding** scale factor so the signal is comparable over time

Exact baseline formula:

```
raw = volume_this_minute / rolling_mean(volume, window=30, min_periods=5)

# Scale factor: expanding mean of |raw| over all history so far,
# requiring at least 30 bars of raw before the first valid value,
# then lagged by 1 bar (no look-ahead).
long_run_scale = expanding_mean(|raw|, min_periods=30).shift(1)

signal = raw / long_run_scale
```

Notes on the two different "30"s:

- **30 in** `rolling_mean` — local lookback: "busy vs the last half hour"

- **30 in** `expanding_mean(..., min_periods=30)` — warm-up only: need ~30 one-minute `raw` observations before the scale is defined; after that the denominator uses **all** past `|raw|` history (expanding), not another rolling window and not "30 blocks of 30 minutes"

Elevated values indicate unusually high activity versus recent history, after rescaling by how large surges typically have been. Trade direction typically follows the signed, position-sized form of the signal.

Volume surge measures **activity intensity**, not price momentum. Momentum may be studied later as a complementary feature; it is not the focus of this ticket.

Metrics glossary

Term	Definition
Forward return	GLD return after signal time (e.g. next 1 / 5 / 10 / 30 minutes)
bps (basis points)	1 bp = 0.01%; avg bps/trade = mean trade P&L in that unit
Hit rate	Fraction of profitable trades (often near 50% even for useful signals)
Sharpe	Return per unit of total volatility; compare only like-for-like constructions
Sortino	Return per unit of downside volatility (penalizes losses, not upside variance); report alongside Sharpe
IC (information coefficient)	Correlation of signal vs forward return; ranking diagnostic, not live P&L
Risk-capped / position-sized	Map signal to a bounded position (e.g. rolling z-score with clip)
Non-overlapping holds	No new entry until the prior hold completes
OOS / walk-forward	Evaluation on periods not used for specification choice
TOD	Time of day
Lead/lag	Relative timing between GC, MGC, and GLD (in minutes)
Promote / monitor / kill	Advance toward live research / watch / discontinue

Scope

Compare volume-surge specifications for short-horizon GLD prediction across five axes:

1. **Lookback window** — history length that defines "normal" volume
2. **Volume ratio construction** — mean, median, percentile, log-ratio, and related forms
3. **Velocity / acceleration** — whether the burst is building, peaking, or fading
4. **Time of day** — session windows where each definition is predictive
5. **GC vs MGC** — level differences, timing, and lead/lag between contracts

Research plan

A. Lookback window

The baseline used a **30-minute** trailing average. Sweep alternatives such as:

- 5, 10, 15, 30, 60, 90, 120 minutes

Also evaluate:

- Whether optimal lookback varies by session (open vs close)
- Expanding vs rolling normalization; sensitivity to minimum history before the signal is valid

Decision required: Is 30 minutes structurally preferred, or an artifact of the initial choice?

B. Volume ratio construction

Beyond `volume / trailing_mean`:

- Trailing **median** (more robust to single extreme minutes)
- Percentile rank vs recent history, or vs the same time of day over recent days
- Log-ratio vs linear ratio
- Uptick vs downtick vs total volume surges
- Instrument sets: GC only, MGC only, GC+MGC, ETF basket, futures/ETF cross-ratio

Decision required: Which definition of unusual volume best predicts GLD under fair evaluation?

C. Velocity / acceleration

- Static surge: is volume elevated now?

- Velocity: is volume becoming more unusual?

Examples:

- Change in surge over the last 1 / 5 / 15 minutes
- Duration above a threshold before entry
- Entry on the rising flank vs peak vs post-confirmation

Decision required: Prefer onset, peak, or persistence of the burst?

D. Time of day

Map performance by Eastern session bands, for example:

- Open (09:30–10:00)
- Around the London PM gold fix (~10:00 US Eastern)
- Late morning, lunch, early/mid afternoon
- Early close (15:00–15:30), last half-hour (15:30–16:00), full close hour

For each band, evaluate forward horizons of 1m / 5m / 10m / 30m.

Caution: Large average bps with very few trades (e.g. one trade per day) can be statistically fragile and operationally weak. Always report trade count and stability.

Decision required: Which surge definition belongs in which window, and which windows should be excluded?

E. GC vs MGC differences and lead/lag

Treat GC and MGC as related but distinct clocks:

- Cross-correlate GC vs MGC surge (and raw volume) at lags −N…+N minutes
- By TOD: which contract's surge more often leads GLD?
- Sign-agreement (GC ∩ MGC) vs single-contract legs
- Differentials: GC−MGC or GC/MGC as standalone features
- Data completeness: GC coverage was incomplete in an earlier merge (~87% join). Assess whether MGC can substitute on sparse GC days

Decision required: Recommend GC, MGC, both, or GC primary with MGC backup — including lag structure if relevant.

Method

1. Reproduce the baseline (30-minute GC volume / trailing mean, close window) and reconcile any material differences with prior ballpark results.
2. Vary one axis at a time (lookback → ratio → velocity → TOD → GC/MGC), then combine the strongest candidates.
3. Use a fast screen (IC or simple signed P&L) for ranking; require risk-capped, non-overlapping trade stats for any production recommendation.
4. Prefer walk-forward / held-out evaluation over a single full-sample score.
5. Present results so the desk can choose: retain the current signal, replace it, or run dual sleeves (e.g. open vs close).

Confirm data location and the existing GLD + futures 1-minute merge with a mentor before introducing a new data path.

Deliverables

1. Concise write-up of the recommended surge definition(s), including exact formula
2. Spec table: formula, lookback, instrument (GC/MGC/ETF), TOD window, hold horizon
3. Ranked results: avg bps/trade, Sharpe, Sortino, hit rate, trade count, drawdown (risk-capped)
4. Heatmaps: time of day × horizon for top specifications
5. GC vs MGC note: lead/lag findings and primary/backup recommendation
6. Decision: promote / monitor / kill relative to the current close GC volume-surge candidate
7. Link write-up / notebook / report on this ticket

Acceptance criteria

- Baseline 30-minute GC surge reproduced, or differences explained
- Lookback and ratio sweeps completed; winners documented by time of day
- Velocity/acceleration variants tested against the static baseline
- GC vs MGC lead/lag quantified with a clear recommendation
- Risk-capped stats for finalists, including Sharpe, Sortino, and trade counts
- Sparse high-bps setups assessed without overstating capacity
- Ticket summary readable without prior Strat packaging knowledge

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Definition of done

The assignee can state, with evidence:

> The recommended volume-surge definition; the session windows where it applies; whether GC or MGC leads; and whether it improves on the prior 30-minute close specification on a fair, trade-count-aware basis.

If no specification beats the baseline, document that outcome and stop expanding the family.

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